A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Global climate change - Set 2

Mark scheme

AQA spec 3.2.1.2; The physical environment; textbook pp. 89-107

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**Any 3 of 6** 1 mark each for any 3 of the 6 points below, in any order, maximum 3. **1** Raise river banks / build sea walls, flood barriers or wave screens **1** Permeable urban surfaces (gravel, soil) instead of concrete / tarmac to slow runoff **1 River flow management:** upstream dams / low soil dams / afforestation / maintained flood plains to delay runoff **1** Raised buildings on stilts or floating houses **1** Pumping rainwater from land below river level **1** Managed retreat - abandoning land where protection costs exceed value (eg Happisburgh) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Permeable** → how easily water passes through / lets water through **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3

Do not write outside the box

Question	Marking guidance	Marks
01.2	**1 mark each** 1 mark for each of the 2 points below. **1** Where the cost of flood prevention is greater than the value of what is protected, or defences would be ineffective **1** Land is abandoned as water levels rise / coast erodes – defences used only where land is very important **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
01.3	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Road surfaces deform as tar melts - need tarmac with a higher melting point **1** Rail track expands and buckles above the temperature it was pre-stretched for (27 °C in UK) - relay with higher pre-stress **1** Bridge damage from high river flows / debris – landslides on deforested slopes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
01.4	**1 mark each** 1 mark for each of the 3 points below. **1** CO_2 is captured from the flue gases (or carbon removed from the fuel before combustion) **1** Transported by pipeline / ship / tanker and stored in depleted oil or gas fields, aquifers, or used for secondary oil recovery **1 Limitation:** developmental / expensive, uses energy (reducing efficiency), risk of leakage, limited storage sites **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **carbon capture** → capturing carbon dioxide from flue gases and storing it underground	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 99-100, 105-108; AQA spec 3.2.1.2.	

Do not write outside the box

Question	Marking guidance	Marks
02.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Control of greenhouse gases:** energy conservation (transport, buildings, industry), renewable and nuclear energy replacing fossil fuels, CCS, reduced deforestation and afforestation (carbon sequestration), methane capture from landfill / digesters, HFC controls, changes in agriculture • **Geoengineering:** white roofs, ocean fertilisation, solar shades - largely untried, possible unpredictable consequences • **International agreements:** Kyoto Protocol, Paris Agreement - targets, but non-binding / uneven participation – carbon trading and taxes – comparison of per capita carbon footprints between countries • **Evidence of limited success:** global CO2 concentrations still rising (about 280 ppm pre-industrial to over 410 ppm) – temperatures up about 1 °C – emissions in some developed countries falling but rising in developing economies • **Reasons for difficulty:** fossil fuels underpin economies, need for major lifestyle change, cost, time delays and natural fluctuations masking effects, political disagreement, developing countries' right to develop • Adaptation strategies (flood defences, drainage, managed retreat, infrastructure changes) reduce impacts but do not address the cause • Comparison with the Montreal Protocol's success for ozone • **Judgement:** partial success - technology exists and some regional reductions achieved, but global strategies have not yet stopped the rise – tipping points make urgency greater **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	9
	• Any accurate, relevant material that is not in the indicative content. • **energy conservation** → using less energy, eg by insulation or behaviour change • **CCS** → capturing carbon dioxide from flue gases and storing it underground **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	

Do not write outside the box

Question	Marking guidance	Marks
	Model answer - Level 3 (8 marks)	
	Strategies fall into control of greenhouse gases, geoengineering, international agreements and adaptation. Emissions are cut by energy conservation in transport, buildings and industry, by replacing fossil fuels with renewables and nuclear, by carbon capture and storage, by reducing deforestation and planting forests, and by capturing methane from landfill. Geoengineering ideas - white roofs, ocean fertilisation, solar shades - remain largely untried with unpredictable side effects. Adaptation, such as flood defences and managed retreat, reduces impacts but not the cause.	
	The Kyoto Protocol and Paris Agreement set targets, but they are weak or non-binding and participation is uneven; carbon trading and taxes exist but per capita footprints still differ enormously between countries. The evidence of success is limited: atmospheric CO_2 has risen from about 280 ppm before industrialisation to over 410 ppm and temperature by about 1 °C, and although emissions are falling in some developed countries they are rising in developing economies.	
	The difficulty is that fossil fuels underpin economies, so control demands lifestyle change, cost, and political agreement between nations with different priorities, while time delays and natural fluctuations mask the effect of any action. By contrast the Montreal Protocol succeeded because ozone was a substitution problem. The judgement must be partial success: the technology exists and regional reductions prove it works, but global strategies have not yet stopped the rise, and tipping points make the delay dangerous.	
	Weaker answer - Level 2 (5 marks)	
	Methods to reduce climate change include using renewable energy such as wind and solar instead of fossil fuels, saving energy, and planting trees to absorb carbon dioxide. International agreements like the Kyoto Protocol and the Paris Agreement have set targets for countries to reduce emissions. However carbon dioxide levels are still rising, so the methods have not been very successful so far. This is because countries rely on fossil fuels and do not always keep to the agreements. More needs to be done.	
	What makes the difference	
	• Level 2 covers the main strategies and reaches a conclusion, but with no quantitative evidence (280 to 410 ppm, 1 °C) and no mention of geoengineering or adaptation.	
	• Level 3 explains *why* success is limited (economic dependence, time lags, natural variability, political disagreement) rather than just stating that it is.	
	• The Montreal comparison and the tipping-point point give Level 3 the breadth and judgement the top band requires.	
02.2	1 mark for the point below. **1** The total greenhouse gas / CO2 emissions caused by an individual, organisation, product or country (often expressed per capita per year) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 89-107; AQA spec 3.2.1.2.	
03.1	**1 mark each** 1 mark for each of the 3 points below. **1** 2020 has higher peak flow in the wet season (32 vs 22 $m^3 s^{-1}$ in March) **1** 2020 has much lower dry-season flow (2 vs 9 $m^3 s^{-1}$ in July / August) **1** Flow is much more variable / seasonal in 2020 – the river almost dries up **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	3
03.2	**1 mark each** 1 mark for each of the 3 points below. **1** In 1990 snow / ice stored wet-season precipitation and released it gradually as meltwater during the dry season, evening out the flow **1** With less ice, precipitation now falls as rain / melts immediately and runs off quickly, increasing wet-season flow and flood risk **1** Little stored ice remains to melt in the dry season so flow falls very low **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	3
03.3	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Water shortages for irrigation / drinking / HEP in the dry season **1** Flooding in the wet season **1** Sudden floods from ice-lake outbursts (glacial lake outburst floods) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **HEP** → electricity generated from falling water **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2
03.4	**1 mark each** 1 mark for each of the 2 points below. **1** Surface meltwater flows down cracks to the base, lubricating the ice so it slides faster – if this exceeds the faster melting at the front, the front advances **1** But if snowfall at the source has not increased, total volume falls – both retreating and extending glaciers can indicate warming **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 94-96; AQA spec 3.2.1.2.	

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